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Sam Schlesinger

Professional Experience

2022–Present **Senior Software Engineer**, *Google Chrome AI Security*

- Implemented a private zero-knowledge banking protocol.
- Co-authored the Anonymous Tokens with Hidden Metadata standards drafts and Rust implementation; deployed to protect billions of users from spam texts via RCS/iMessage integration.
- Designed and implemented the Private Verification Tokens API in Chromium using ATHM to transfer low-entropy trust from regular to private browsing, reducing CAPTCHA friction while preserving privacy.
- Co-authored the Anonymous Credit Tokens (ACT) standards drafts and Rust implementation for privacy-preserving API credits; presented ACT to the IETF Privacy Pass Working Group at IETF 123, IETF 124, and its May 2026 interim.
- Led a three-day cross-browser PACT workshop producing three Moderation of unLinkable Endorsements (MoLE) Internet-Drafts; built an end-to-end Chromium prototype; presented at IETF 126. BoF expected at IETF 127 in San Francisco.
- Designed the Private Proof API for anti-fraud in a post-third-party-cookie web. Led three engineers to implement in Chrome. Drafted novel anonymous credential primitives.
- Co-chair of W3C Anti-Fraud Community Group.

2022 **Software Engineer / Scientist**, *Casper Labs*

- Co-designed novel consensus algorithm (presented at FOCODILE 2022). Implemented in Casper node, deployed in 2.0 upgrade.
- Researched zero-knowledge use cases and proposed network optimizations.
- Consulted C-suite on network security and scalability strategy.

2019–2022 **Software Engineer (Team Lead)**, *SimSpace*

- Designed and implemented Attack Designer program.
- Designed consensus-lite protocol for multi-datacenter cluster coordination.
- Backend security lead: identified and fixed cryptographic vulnerabilities.

Community & Outreach

Creator Computable Secrets, a YouTube channel explaining cryptography, formal methods, and theoretical computer science.

Reviewer CSLib, reviewing formalized computer science contributions to its Lean library.

Instructor Delivered tailored computer science lectures at local high schools and taught Google Code Next classes to high school students.

Skills

Core Cryptography, Security, Distributed Systems, Internet Standards

Engineering Rust, Haskell, TypeScript/JavaScript, Web Servers, PostgreSQL, Kubernetes, Smart Contracts

Education

2019 **B.A. Computer Science and Math**, *UMass*, Amherst